# VetCare Herts: a synthetic relational database for a veterinary practice group

Data Mining and Discovery: SQL Database Project | Omkumar Patel | Student ID: 24174049

*Code: https://github.com/Patel-om13/Data_mining_and_Discovery.git | Database: vetcare_clinics.db*

## 1. Purpose of the database

In this project, I created a database for an imaginary veterinary practice called VetCare Herts, comprising of 7 veterinary practices in Hertfordshire and Bedfordshire. It has the clinics, the vets that work there, the weekly rotation of vets, the clients and their animals, bookings between January 2024 and September 2026 and the treatments charged at each booking. I chose this topic as it is the kind of data that is generated in a vet practice: species, ordered triage scale, temperatures, dates, money, weights. A practice manager could utilize it to understand which clinics are the busiest, and the number of missed appointments and where the revenue is from. This file contains 8 tables with 35,042 rows. The two biggest are appointment_treatments (20,574 rows) and appointments (10,210 rows).

## 2. How the data was generated

All this was created in-house. Each distribution (row) is generated within my own Python notebook, using NumPy (Harris et al., 2020), with British names and towns provided by Faker running with en_GB (Faker, 2026). The seed is set to 2026, and the model is built randomly in each run, so the model will be the same database. The tables are set up first by running the DDL script, then the data is inserted in the parent order: clinics, vets, rota, owners, pets, treatments, appointments and finally the treatment lines. Foreign key enforcement is enabled before loading, so that SQLite won't accept any row that refers to a missing parent (SQLite, no date a).

Most of the work was put into making the values depend on and not drawing each column independently. An owner's email is made up of their own name with a weighted choice of UK providers, mobiles use the 07xxx format, and postcode districts are from the area around the owner's clinic. The weights of pets are normally distributed and breed specific; the mean weight of a Labrador is about 32 kg, while that of a budgerigar is about 35 g and the weights of pets under 1 year of age are less. Body temperature varies by species and a few sickness or emergency calls are assigned a fever. Seasonal, and weekday weights are assigned to the appointment dates and with Saturday being the busiest, and Sunday only being covered at the two 24 hour facilities. Vets can only be booked on days in the rota when the vet is in the office. The length of the visit is generated from a gamma distribution for each reason and the fee is based on the total of the treatment lines plus an annual price increase of five per cent, which is an average completed visit of £173.07. Approximately 6% of bookings are canceled, and 5% are no shows.

Real practice systems always get dirty, so I got a copy of the clean tables dirty myself, in one obviously practice notebook step. Each of these error rates is a constant I have selected and the IDs that I modified are printed, and subsequent SQL cells keep tabs on each one of them.

| Issue | Where | Rate | What it imitates |
|---|---|---|---|
| Missing | owners.mobile / owners.email | 4% / 2.5% (65 mobiles) | Client declined, or an old paper record |
| Missing | pets.weight_kg | 3%, 8% for birds and reptiles (75) | Animal not weighed |
| Missing | appointments.body_temp_c, payment_method | 3% / 1.5% of completed visits | Reading not taken; payment not reconciled |
| Duplicate | owners | 1.5% (24 records) | Client registered twice with a case or spacing change |
| Duplicate | appointments | 0.5% (51 records) | Booking saved twice |

## 3. Database schema

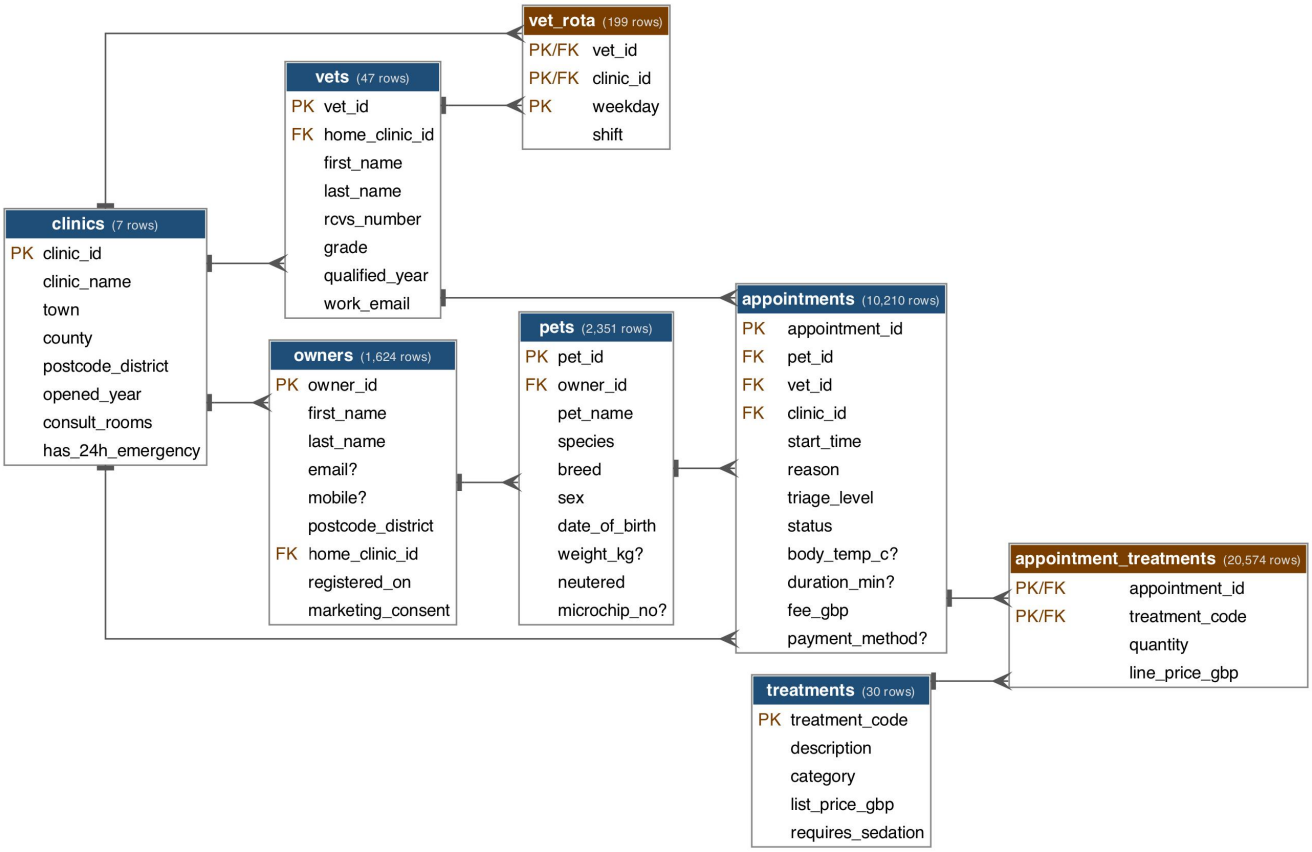



*Figure 1. VetCare Herts tables and their links, read directly out of vetcare_clinics.db. Crow's feet mark the "many" side, "?" marks a nullable column and the brown headers are the two tables with composite primary keys.*

## 4. Justification of tables and constraints

I followed Codd's (1970) rules and split the data down to 3NF (3rd normal form), where each fact is stored in only one location. Once the owner of three dogs appears as a client, a change in the price is made to one row of treatment. I assigned a storage type to all 60 columns, then added at least one NOT NULL, CHECK, UNIQUE or key rule to each of these (SQLite, no date b), while foreign keys (10 of them) are defined with ON DELETE rules, akin to what a practice would do: removing an owner removes their pets; but if a vet has pets in the appointment table, then they cannot be removed.

| Table (rows) | Why it is a separate table | Main constraints |
|---|---|---|
| clinics (7) | Reference data shared by vets, owners and appointments. | UNIQUE name; county IN list; postcode length and letter check; ranges on opened_year and consult_rooms |
| vets (47) | Staff details change independently of appointments. | FK home clinic (RESTRICT); grade IN four ordered grades; 7-digit RCVS number UNIQUE; email UNIQUE |
| vet_rota (199) | Resolves the many-to-many link between vets and clinics on each weekday. | Composite PK (vet_id, clinic_id, weekday); weekday 1 to 7; shift IN list; CASCADE on delete |
| owners (1624) | Personal data kept apart from clinical data so access can be limited. | Email needs text before an @ and a dot after it; mobile is 12 characters, digits and one space (both nullable); ISO date CHECK; consent 0 or 1 |
| pets (2351) | One owner can have many animals. | FK owner (CASCADE); species IN list; weight 0.01 to 110 kg; 15-digit microchip UNIQUE; composite UNIQUE (owner, name, birth date) |
| treatments (30) | Price list and lookup, so descriptions are not repeated. | Code GLOB AAA-000; category IN list; price at least 0 |
| appointments (10,210) | The central event table linking pet, vet and clinic. | Three FKs; reason and status IN lists; triage 1 to 5; temperature 35 to 43.5; duration 5 to 480; fee at least 0 |
| appointment_treatments (20,574) | Resolves the many-to-many link between visits and treatments. | Composite PK (appointment_id, treatment_code); quantity 1 to 20; price at least 0 |

There are two options that should be explained. I intentionally did NOT put the UNIQUE constraint into owners.email as it would not accept the duplicate registrations I wanted to retain for cleaning. Then I would add a UNIQUE index with NOCASE collation. I also only accepted a NULL value where it might be null. For example, the body temperature of reptiles is never taken since they are ectothermic.

I checked my columns against the scale types in Stevens (1946), and each type turns up at least twice:

| Level | Columns | Reason |
|---|---|---|
| Nominal | species, breed, reason, payment_method, county | Labels with no natural order |
| Ordinal | triage_level (1 routine to 5 critical), vets.grade | Ordered, but the gaps are not equal |
| Interval | body_temp_c, start_time, date_of_birth, opened_year | Equal steps but an arbitrary zero (0 °C, year 0) |
| Ratio | weight_kg, fee_gbp, duration_min, quantity | A true zero, so "twice as heavy" makes sense |

## 5. Ethics and data privacy

All the people in the database are fictional, and no real person can be harmed by it. That's why synthetic data is helpful in constructing and testing a system before it contains any real client data. A generated name might still be someone who does exist, so I added random contact information to each name and never would I use these records as real clients. The vets' RCVS numbers are also arbitrary and do not correspond to the actual numbers.

If the same design contained actual data, it would fall under the scope of the UK GDPR and the Data Protection Act 2018, as names, emails and mobile numbers are personal data. I adhered to the data minimisation principle (ICO, no date a) Only half of the postcode is kept, no one has a complete address or a date of birth, and details of cards are not kept. Marketing consent is its own space and if a customer does not wish to be contacted they are not. The clinical tables assign identifiers to people instead of using their name, allowing an analyst to study visits without exposing names. But it is not anonymisation (ICO, no date b), it is pseudonymisation, as the ID can be related back to the owner. For this reason, owners table should be behind role based access, file should be encrypted at rest and records deleted or anonymised after a certain retention period. I also recorded the injected errors in the notebook, just in case someone thinks that this is a data quality error.

## 6. Validation and example use

After loading, PRAGMA integrity_check returned "ok" and PRAGMA foreign_key_check found 0 violations. Grouping owners on lower-case, trimmed names plus mobile brings back the 24 injected duplicates. Grouping appointments on pet, vet and start time finds the 51 double bookings. The query below joins three tables to show the kind of question the database can answer.

```
SELECT t.category, SUM(at.quantity) AS units, ROUND(SUM(at.line_price_gbp)) AS revenue_gbp
FROM appointment_treatments at JOIN treatments t USING (treatment_code)
JOIN appointments a USING (appointment_id) WHERE a.status = 'Completed'
GROUP BY t.category ORDER BY revenue_gbp DESC;
```

| Category | Units | Revenue (£) |
|---|---|---|
| Consultation | 9,117 | 474,302 |
| Diagnostics | 2,309 | 301,472 |
| Surgery | 839 | 263,127 |
| Medication | 7,068 | 208,247 |
| Dental | 912 | 176,418 |
| Vaccination | 2,308 | 135,170 |
| Parasite control | 775 | 14,001 |

## References


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## Appendix A:

I wanted microchips to behave like the real register, so only dogs and cats get one, at different rates, and no number may appear twice. These lines from generate_vetcare.ipynb draw a 15-digit code and redraw it on any clash.

```
chip = None
if sp in ('Dog','Cat') and rng.random() < (.97 if sp == 'Dog' else .78):  # dog chipping is a UK legal duty
    chip = '9' + ''.join(map(str, rng.integers(0, 10, 14)))
    while chip in chips:
        chip = '9' + ''.join(map(str, rng.integers(0, 10, 14)))
    chips.add(chip)
```